Single_Wrapper_Test

May 31, 2026

[7]:
```
pip install --upgrade gotabpfn
```

```
Requirement already satisfied: gotabpfn in /usr/local/lib/python3.12/dist-
packages (0.1.10)
Collecting gotabpfn
  Downloading gotabpfn-0.1.11-py3-none-any.whl.metadata (116 kB)
                           116.1/116.1

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Requirement already satisfied: numpy>=1.23 in
/usr/local/lib/python3.12/dist-packages (from gotabpfn) (2.0.2)
Requirement already satisfied: pandas>=1.5 in /usr/local/lib/python3.12/dist-
packages (from gotabpfn) (2.2.2)
Requirement already satisfied: scipy>=1.11 in /usr/local/lib/python3.12/dist-
packages (from gotabpfn) (1.16.3)
Requirement already satisfied: scikit-learn>=1.2 in
/usr/local/lib/python3.12/dist-packages (from gotabpfn) (1.6.1)
Requirement already satisfied: tqdm>=4.64 in /usr/local/lib/python3.12/dist-
packages (from gotabpfn) (4.67.3)
Requirement already satisfied: optuna>=3.5 in /usr/local/lib/python3.12/dist-
packages (from gotabpfn) (4.8.0)
Requirement already satisfied: torch>=2.1 in /usr/local/lib/python3.12/dist-
packages (from gotabpfn) (2.11.0+cu128)
Requirement already satisfied: tabpfn==6.3.1 in /usr/local/lib/python3.12/dist-
packages (from gotabpfn) (6.3.1)
Requirement already satisfied: kmeans-gpu==0.0.5 in
/usr/local/lib/python3.12/dist-packages (from gotabpfn) (0.0.5)
Requirement already satisfied: matplotlib>=3.7 in
/usr/local/lib/python3.12/dist-packages (from gotabpfn) (3.10.0)
Requirement already satisfied: typing_extensions>=4.12.0 in
/usr/local/lib/python3.12/dist-packages (from tabpfn==6.3.1->gotabpfn) (4.15.0)
Requirement already satisfied: einops<0.9,>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from tabpfn==6.3.1->gotabpfn) (0.8.2)
Requirement already satisfied: huggingface-hub<2,>=0.19.0 in
/usr/local/lib/python3.12/dist-packages (from tabpfn==6.3.1->gotabpfn) (1.16.1)
Requirement already satisfied: pydantic>=2.8.0 in
/usr/local/lib/python3.12/dist-packages (from tabpfn==6.3.1->gotabpfn) (2.12.3)
Requirement already satisfied: pydantic-settings>=2.10.1 in
/usr/local/lib/python3.12/dist-packages (from tabpfn==6.3.1->gotabpfn) (2.14.1)
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Requirement already satisfied: eval-type-backport>=0.2.2 in
/usr/local/lib/python3.12/dist-packages (from tabpfn==6.3.1->gotabpfn) (0.3.1)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-
packages (from tabpfn==6.3.1->gotabpfn) (1.5.3)
Requirement already satisfied: tabpfn-common-utils>=0.2.13 in
/usr/local/lib/python3.12/dist-packages (from tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (0.2.21)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7->gotabpfn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7->gotabpfn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7->gotabpfn)
(4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7->gotabpfn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7->gotabpfn) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7->gotabpfn) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7->gotabpfn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7->gotabpfn)
(2.9.0.post0)
Requirement already satisfied: alembic>=1.5.0 in /usr/local/lib/python3.12/dist-
packages (from optuna>=3.5->gotabpfn) (1.18.4)
Requirement already satisfied: colorlog in /usr/local/lib/python3.12/dist-
packages (from optuna>=3.5->gotabpfn) (6.10.1)
Requirement already satisfied: sqlalchemy>=1.4.2 in
/usr/local/lib/python3.12/dist-packages (from optuna>=3.5->gotabpfn) (2.0.49)
Requirement already satisfied: PyYAML in /usr/local/lib/python3.12/dist-packages
(from optuna>=3.5->gotabpfn) (6.0.3)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5->gotabpfn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5->gotabpfn) (2026.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.2->gotabpfn)
(3.6.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-
packages (from torch>=2.1->gotabpfn) (3.29.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-
packages (from torch>=2.1->gotabpfn) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-
packages (from torch>=2.1->gotabpfn) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.1->gotabpfn) (3.6.1)
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Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
(from torch>=2.1->gotabpfn) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-
packages (from torch>=2.1->gotabpfn) (2025.3.0)
Requirement already satisfied: cuda-toolkit==12.8.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.1->gotabpfn) (12.9.6)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.1->gotabpfn) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.1->gotabpfn) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.1->gotabpfn) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.1->gotabpfn) (3.4.5)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-
packages (from torch>=2.1->gotabpfn) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (11.7.3.90)
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Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.5.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.93)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.93)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=2.1->gotabpfn) (12.8.90)
Requirement already satisfied: Mako in /usr/local/lib/python3.12/dist-packages
(from alembic>=1.5.0->optuna>=3.5->gotabpfn) (1.3.12)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-
bindings<13,>=12.9.4->torch>=2.1->gotabpfn) (1.5.4)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (1.5.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (0.28.1)
Requirement already satisfied: typer>=0.20.0 in /usr/local/lib/python3.12/dist-
packages (from huggingface-hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (0.25.1)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from
pydantic>=2.8.0->tabpfn==6.3.1->gotabpfn) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in
/usr/local/lib/python3.12/dist-packages (from
pydantic>=2.8.0->tabpfn==6.3.1->gotabpfn) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in
/usr/local/lib/python3.12/dist-packages (from
pydantic>=2.8.0->tabpfn==6.3.1->gotabpfn) (0.4.2)
Requirement already satisfied: python-dotenv>=0.21.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic-
settings>=2.10.1->tabpfn==6.3.1->gotabpfn) (1.2.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.7->gotabpfn) (1.17.0)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-
packages (from sqlalchemy>=1.4.2->optuna>=3.5->gotabpfn) (3.5.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from
sympy>=1.13.3->torch>=2.1->gotabpfn) (1.3.0)
```

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Requirement already satisfied: nvidia-ml-py>=13.590.48 in
/usr/local/lib/python3.12/dist-packages (from tabpfn-common-
utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (13.595.45)
Requirement already satisfied: platformdirs>=4 in
/usr/local/lib/python3.12/dist-packages (from tabpfn-common-
utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (4.9.6)
Requirement already satisfied: posthog>=6.7 in /usr/local/lib/python3.12/dist-
packages (from tabpfn-common-utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (7.16.2)
Requirement already satisfied: requests>=2.32.5 in
/usr/local/lib/python3.12/dist-packages (from tabpfn-common-
utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (2.34.2)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=2.1->gotabpfn)
(3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
(from httpx<1,>=0.23.0->huggingface-hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn)
(4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (2026.5.20)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages
(from httpx<1,>=0.23.0->huggingface-hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn)
(3.15)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-
packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (0.16.0)
Requirement already satisfied: backoff>=1.10.0 in
/usr/local/lib/python3.12/dist-packages (from posthog>=6.7->tabpfn-common-
utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (2.2.1)
Requirement already satisfied: distro>=1.5.0 in /usr/local/lib/python3.12/dist-
packages (from posthog>=6.7->tabpfn-common-utils>=0.2.13->tabpfn-common-
utils[telemetry-interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (1.9.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.5->tabpfn-common-
utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.26 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.5->tabpfn-common-
utils>=0.2.13->tabpfn-common-utils[telemetry-
interactive]>=0.2.13->tabpfn==6.3.1->gotabpfn) (2.5.0)
```

```
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-
packages (from typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (8.4.0)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-
packages (from typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from
rich>=13.8.0->typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from
rich>=13.8.0->typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer>=0.20.0->huggingface-
hub<2,>=0.19.0->tabpfn==6.3.1->gotabpfn) (0.1.2)
Downloading gotabpfn-0.1.11-py3-none-any.whl (84 kB)
                         84.7/84.7 kB
3.7 MB/s eta 0:00:00
Installing collected packages: gotabpfn
  Attempting uninstall: gotabpfn
    Found existing installation: gotabpfn 0.1.10
    Uninstalling gotabpfn-0.1.10:
      Successfully uninstalled gotabpfn-0.1.10
Successfully installed gotabpfn-0.1.11
```

[1]:
```
pip show gotabpfn
```

```
Name: gotabpfn
Version: 0.1.11
Summary: GOTabPFN: From Feature Ordering to Compact Tokenization for Tabular
Foundation Models on High-Dimensional Data
Home-page: https://github.com/zadid6pretam/GOTabPFN
Author: Al Zadid Sultan Bin Habib, Md Younus Ahamed, Prashnna Kumar Gyawali,
Gianfranco Doretto, Donald A. Adjeroh
Author-email:
License: MIT
Location: /usr/local/lib/python3.12/dist-packages
Requires: kmeans-gpu, matplotlib, numpy, optuna, pandas, scikit-learn, scipy,
tabpfn, torch, tqdm
Required-by:
```

1 DrivFace - Regression || Single Wrapper

[5]:
```python
import numpy as np
import torch

from gotabpfn import run_gotabpfn_csv


# ------------------------
# User settings
# ------------------------
DATA_FILE = "drivface.csv"  # change this to your dataset file name
TARGET_COL = "angle"        # change this to your regression target column
SEED = 42

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"


# ------------------------
# Fixed GO-LR hyperparameters
# ------------------------
GO_METRIC = "manhattan"
GO_NUM_CLUSTERS = 5
GO_REFINE_PASSES = 1
GO_DIRECTION_SELECT = False
GO_FEAT_SUBSAMPLE = 2000


# ------------------------
# Fixed NSC-pSP hyperparameters
# ------------------------
NSC_SEGMENTATION = "largest_jump"
NSC_M_RULE = "idf"
NSC_TAU = 0.99
NSC_GAMMA = 2.654390393837633
NSC_BETA = 0.043192175152615336
NSC_MMIN = 16
NSC_MMAX = 256
NSC_LMIN = 12
ASSUME_STANDARDIZED = True

TABPFN_SEED = 3


# ------------------------
# Run GOTabPFN
# ------------------------
```

```python
results = run_gotabpfn_csv(
    csv_path=DATA_FILE,
    target_col=TARGET_COL,
    task_type="regression",
    non_numeric="drop",

    # 5x5 repeated cross-validation
    cv="5x5",
    seed=SEED,

    # GO-LR settings
    go_metric=GO_METRIC,
    go_num_clusters=GO_NUM_CLUSTERS,
    go_refine=True,
    go_direction_select=GO_DIRECTION_SELECT,
    go_refine_passes=GO_REFINE_PASSES,
    go_feat_subsample=GO_FEAT_SUBSAMPLE,

    # Try GPU KMeans first, then fall back to CPU KMeans if needed
    go_use_cpu_kmeans=False,
    go_fallback_cpu_kmeans=True,

    # NSC-pSP settings
    nsc_segmentation=NSC_SEGMENTATION,
    nsc_m_rule=NSC_M_RULE,
    nsc_tau=NSC_TAU,
    nsc_gamma=NSC_GAMMA,
    nsc_beta=NSC_BETA,
    nsc_M_min=NSC_MMIN,
    nsc_M_max=NSC_MMAX,
    nsc_l_min=NSC_LMIN,
    assume_standardized=ASSUME_STANDARDIZED,

    # TabPFN-2.5 head
    tabpfn_seed=TABPFN_SEED,
    device=DEVICE,

    return_predictions=True,
    verbose=False,
)


# -----------------------
# Access outputs
# -----------------------
print(results["summary"])
```

```python
metrics_df = results["metrics_df"]
predictions_df = results["predictions_df"]

print("\nFold-wise metrics")
print(metrics_df)

print("\nPrediction preview")
print(predictions_df.head())

print("\nGO-LR ordering")
print(f"Learned GO-LR order length: {len(results['ordering'])}")
print("First 10 ordered features:")
print(results["ordered_feature_names"][:10])


# ------------------------
# Final 5x5 CV summary
# ------------------------
r2s = metrics_df["r2"].to_numpy()
rmses = metrics_df["rmse"].to_numpy()
maes = metrics_df["mae"].to_numpy()

print("\nFinal 5x5 CV results")
print(f"R2   : {np.mean(r2s):.4f} ± {np.std(r2s, ddof=1):.4f}")
print(f"RMSE : {np.mean(rmses):.4f} ± {np.std(rmses, ddof=1):.4f}")
print(f"MAE  : {np.mean(maes):.4f} ± {np.std(maes, ddof=1):.4f}")
```

```
GOTabPFN completed successfully.

Task type: regression
Evaluation: 5x5 repeated CV
Samples: 606
Original numeric features: 6400
GO-LR metric: manhattan
GO-LR KMeans mode: gpu
GO-LR feature subsample: 2000
NSC segmentation: largest_jump
NSC M rule: idf
NSC bypass_if_m_leq: 0
Device: cuda
Elapsed time: 132.81 seconds

Metric summary:
r2: 0.6548 ± 0.0995
rmse: 8.2695 ± 1.4722
mae: 3.6689 ± 0.6156
nsc_tokens: 256.0000 ± 0.0000
```

```
Fold-wise metrics
    fold        r2       rmse       mae  nsc_tokens
0      1  0.512414  10.792954  4.808915         256
1      2  0.702385   6.701759  2.874984         256
2      3  0.767160   7.359853  3.626536         256
3      4  0.743784   7.608474  3.411266         256
4      5  0.631153   7.991230  3.200625         256
5      6  0.740289   7.367319  3.588372         256
6      7  0.649674   9.099552  4.290849         256
7      8  0.571329   8.833177  3.676428         256
8      9  0.497356   9.076891  3.413999         256
9     10  0.748292   7.312588  3.523059         256
10    11  0.622624   9.356255  3.803633         256
11    12  0.595724   9.949636  3.920303         256
12    13  0.679390   7.707831  3.763957         256
13    14  0.563509   8.462486  3.540481         256
14    15  0.749710   6.904514  3.319380         256
15    16  0.745994   7.227481  3.498031         256
16    17  0.630782   8.408267  3.615018         256
17    18  0.829010   6.256423  2.766467         256
18    19  0.552532   9.662133  4.790599         256
19    20  0.592430   8.572208  3.927465         256
20    21  0.496594  10.328049  3.836538         256
21    22  0.538694  11.446564  5.281747         256
22    23  0.808876   5.226996  2.634352         256
23    24  0.743347   7.090323  3.176280         256
24    25  0.657169   7.994880  3.433472         256

Prediction preview
   row_index  fold  true_value  predicted_value
0          2     1         0.0        -2.887695
1          6     1         0.0        -0.809608
2         10     1         0.0        -0.381828
3         11     1         0.0         0.447103
4         24     1       -15.0         8.835121

GO-LR ordering
Learned GO-LR order length: 6400
First 10 ordered features:
['pixel_1872', 'pixel_1871', 'pixel_1714', 'pixel_1715', 'pixel_1636',
'pixel_1634', 'pixel_2133', 'pixel_2052', 'pixel_2213', 'pixel_1793']

Final 5x5 CV results
R2   : 0.6548 ± 0.0995
RMSE : 8.2695 ± 1.4722
MAE  : 3.6689 ± 0.6156
```

2 orlaws10P || Multiclass Classification || Single Wrapper

[6]:
```
import numpy as np
import torch

from gotabpfn import run_gotabpfn_csv


# ------------------------
# User settings
# ------------------------
DATA_FILE = "orlraws10P.csv"  # change this to your dataset file name
TARGET_COL = "label"          # change this to your target column
SEED = 42

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"


# ------------------------
# Fixed GO-LR hyperparameters
# ------------------------
GO_METRIC = "cosine"
GO_NUM_CLUSTERS = 5
GO_REFINE_PASSES = 1
GO_DIRECTION_SELECT = False
GO_FEAT_SUBSAMPLE = 3000


# ------------------------
# Fixed NSC-pSP hyperparameters
# ------------------------
NSC_SEGMENTATION = "uniform"
NSC_M_RULE = "default"
NSC_TAU = 0.99
NSC_GAMMA = 2.049512863264476
NSC_BETA = 0.3887505167779042
NSC_MMIN = 32
NSC_MMAX = 384
NSC_LMIN = 12
ASSUME_STANDARDIZED = False

TABPFN_SEED = 42


# ------------------------
# Run GOTabPFN
# ------------------------
```

```python
results = run_gotabpfn_csv(
    csv_path=DATA_FILE,
    target_col=TARGET_COL,
    task_type="multiclass",
    non_numeric="drop",

    # 5x5 repeated stratified cross-validation
    cv="5x5",
    seed=SEED,

    # GO-LR settings
    go_metric=GO_METRIC,
    go_num_clusters=GO_NUM_CLUSTERS,
    go_refine=True,
    go_direction_select=GO_DIRECTION_SELECT,
    go_refine_passes=GO_REFINE_PASSES,
    go_feat_subsample=GO_FEAT_SUBSAMPLE,

    # Try GPU KMeans first, then fall back to CPU KMeans if needed
    go_use_cpu_kmeans=False,
    go_fallback_cpu_kmeans=True,

    # NSC-pSP settings
    nsc_segmentation=NSC_SEGMENTATION,
    nsc_m_rule=NSC_M_RULE,
    nsc_tau=NSC_TAU,
    nsc_gamma=NSC_GAMMA,
    nsc_beta=NSC_BETA,
    nsc_M_min=NSC_MMIN,
    nsc_M_max=NSC_MMAX,
    nsc_l_min=NSC_LMIN,
    assume_standardized=ASSUME_STANDARDIZED,

    # TabPFN-2.5 head
    tabpfn_seed=TABPFN_SEED,
    device=DEVICE,

    return_predictions=True,
    verbose=True,
)


# -----------------------
# Access outputs
# -----------------------
print(results["summary"])
```

```python
metrics_df = results["metrics_df"]
predictions_df = results["predictions_df"]

print("\nFold-wise metrics")
print(metrics_df)

print("\nPrediction preview")
print(predictions_df.head())

print("\nGO-LR ordering")
print(f"Learned GO-LR order length: {len(results['ordering'])}")
print("First 10 ordered features:")
print(results["ordered_feature_names"][:10])


# ------------------------
# Final 5x5 CV summary
# ------------------------
accs = metrics_df["accuracy"].to_numpy()
f1s = metrics_df["macro_f1"].to_numpy()

print("\nFinal 5x5 CV results")
print(f"Accuracy      : {np.mean(accs):.4f} ± {np.std(accs, ddof=1):.4f}")
print(f"Macro-F1      : {np.mean(f1s):.4f} ± {np.std(f1s, ddof=1):.4f}")

if "macro_ovr_auc" in metrics_df.columns:
    aucs = metrics_df["macro_ovr_auc"].dropna().to_numpy()
    if len(aucs) > 0:
        print(f"Macro-OvR-AUC : {np.mean(aucs):.4f} ± {np.std(aucs, ddof=1):.
↪4f}")
```

```
X shape: (100, 10304), classes: 10
Using device: cuda
Learned GO-LR order length: 10304
First 10 ordered features:
['feature_339', 'feature_340', 'feature_10195', 'feature_341', 'feature_9633',
'feature_10087', 'feature_116', 'feature_10085', 'feature_10', 'feature_9860']
Fold 01 Z_tr shape: (80, 142)
Fold 01: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 02 Z_tr shape: (80, 142)
Fold 02: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 03 Z_tr shape: (80, 142)
Fold 03: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 04 Z_tr shape: (80, 140)
Fold 04: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 05 Z_tr shape: (80, 142)
Fold 05: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 06 Z_tr shape: (80, 142)
```

```
Fold 06: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 07 Z_tr shape: (80, 142)
Fold 07: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 08 Z_tr shape: (80, 142)
Fold 08: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 09 Z_tr shape: (80, 140)
Fold 09: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 10 Z_tr shape: (80, 142)
Fold 10: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 11 Z_tr shape: (80, 142)
Fold 11: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 12 Z_tr shape: (80, 142)
Fold 12: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 13 Z_tr shape: (80, 142)
Fold 13: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 14 Z_tr shape: (80, 140)
Fold 14: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 15 Z_tr shape: (80, 142)
Fold 15: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 16 Z_tr shape: (80, 142)
Fold 16: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 17 Z_tr shape: (80, 142)
Fold 17: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 18 Z_tr shape: (80, 140)
Fold 18: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 19 Z_tr shape: (80, 142)
Fold 19: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 20 Z_tr shape: (80, 142)
Fold 20: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 21 Z_tr shape: (80, 142)
Fold 21: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 22 Z_tr shape: (80, 142)
Fold 22: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 23 Z_tr shape: (80, 140)
Fold 23: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 24 Z_tr shape: (80, 142)
Fold 24: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000
Fold 25 Z_tr shape: (80, 142)
Fold 25: ACC=1.0000, Macro-F1=1.0000, Macro-OvR-AUC=1.0000

Final CV results
GOTabPFN completed successfully.

Task type: multiclass
Evaluation: 5x5 repeated stratified CV
Samples: 100
Original numeric features: 10304
GO-LR metric: cosine
```

```
GO-LR KMeans mode: gpu
GO-LR feature subsample: 3000
NSC segmentation: uniform
NSC M rule: default
NSC bypass_if_m_leq: 0
Device: cuda
Elapsed time: 47.59 seconds

Metric summary:
accuracy: 1.0000 ± 0.0000
macro_f1: 1.0000 ± 0.0000
macro_ovr_auc: 1.0000 ± 0.0000
nsc_tokens: 141.6000 ± 0.8165
GOTabPFN completed successfully.

Task type: multiclass
Evaluation: 5x5 repeated stratified CV
Samples: 100
Original numeric features: 10304
GO-LR metric: cosine
GO-LR KMeans mode: gpu
GO-LR feature subsample: 3000
NSC segmentation: uniform
NSC M rule: default
NSC bypass_if_m_leq: 0
Device: cuda
Elapsed time: 47.59 seconds

Metric summary:
accuracy: 1.0000 ± 0.0000
macro_f1: 1.0000 ± 0.0000
macro_ovr_auc: 1.0000 ± 0.0000
nsc_tokens: 141.6000 ± 0.8165

Fold-wise metrics
```

	fold	accuracy	macro_f1	macro_ovr_auc	nsc_tokens
0	1	1.0	1.0	1.0	142
1	2	1.0	1.0	1.0	142
2	3	1.0	1.0	1.0	142
3	4	1.0	1.0	1.0	140
4	5	1.0	1.0	1.0	142
5	6	1.0	1.0	1.0	142
6	7	1.0	1.0	1.0	142
7	8	1.0	1.0	1.0	142
8	9	1.0	1.0	1.0	140
9	10	1.0	1.0	1.0	142
10	11	1.0	1.0	1.0	142
11	12	1.0	1.0	1.0	142

```
12     13        1.0        1.0             1.0          142
13     14        1.0        1.0             1.0          140
14     15        1.0        1.0             1.0          142
15     16        1.0        1.0             1.0          142
16     17        1.0        1.0             1.0          142
17     18        1.0        1.0             1.0          140
18     19        1.0        1.0             1.0          142
19     20        1.0        1.0             1.0          142
20     21        1.0        1.0             1.0          142
21     22        1.0        1.0             1.0          142
22     23        1.0        1.0             1.0          140
23     24        1.0        1.0             1.0          142
24     25        1.0        1.0             1.0          142

Prediction preview
   row_index  fold true_label predicted_label
0          1     1          0               0
1          3     1          0               0
2         10     1          1               1
3         11     1          1               1
4         22     1          2               2

GO-LR ordering
Learned GO-LR order length: 10304
First 10 ordered features:
['feature_339', 'feature_340', 'feature_10195', 'feature_341', 'feature_9633',
'feature_10087', 'feature_116', 'feature_10085', 'feature_10', 'feature_9860']

Final 5x5 CV results
Accuracy      : 1.0000 ± 0.0000
Macro-F1      : 1.0000 ± 0.0000
Macro-OvR-AUC : 1.0000 ± 0.0000
```

3 Colon || Binary Classification || Single Wrapper

[2]:
```python
import numpy as np
import torch

from gotabpfn import run_gotabpfn_csv


# -----------------------
# User settings
# -----------------------
DATA_FILE = "coloncancer_encoded.csv"  # change this to your dataset file name
TARGET_COL = "label"                   # change this to your target column
SEED = 42
```

```
# Fixed GO-LR hyperparameters
GO_METRIC = "euclidean"
GO_NUM_CLUSTERS = 10
GO_REFINE_PASSES = 3
GO_DIRECTION_SELECT = True
GO_FEAT_SUBSAMPLE = None

# Fixed NSC-pSP hyperparameters
NSC_SEGMENTATION = "equal_mass"
NSC_M_RULE = "idf"
NSC_TAU = 0.99
NSC_GAMMA = 1.7570143129240916
NSC_BETA = 0.2244046472232107
NSC_MMIN = 64
NSC_MMAX = 384
NSC_LMIN = 16
ASSUME_STANDARDIZED = False

TABPFN_SEED = 42
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"


# -----------------------
# Run GOTabPFN
# -----------------------
results = run_gotabpfn_csv(
    csv_path=DATA_FILE,
    target_col=TARGET_COL,
    task_type="binary",
    non_numeric="drop",

    # 5x5 repeated stratified cross-validation
    cv="5x5",
    seed=SEED,

    # GO-LR settings
    go_metric=GO_METRIC,
    go_num_clusters=GO_NUM_CLUSTERS,
    go_refine=True,
    go_direction_select=GO_DIRECTION_SELECT,
    go_refine_passes=GO_REFINE_PASSES,

    # Try GPU KMeans first, then fall back to CPU KMeans if needed
    go_use_cpu_kmeans=False,
    go_fallback_cpu_kmeans=True,
```

```python
    # NSC-pSP settings
    nsc_segmentation=NSC_SEGMENTATION,
    nsc_m_rule=NSC_M_RULE,
    nsc_tau=NSC_TAU,
    nsc_gamma=NSC_GAMMA,
    nsc_beta=NSC_BETA,
    nsc_M_min=NSC_MMIN,
    nsc_M_max=NSC_MMAX,
    nsc_l_min=NSC_LMIN,
    assume_standardized=ASSUME_STANDARDIZED,

    # TabPFN-2.5 head
    tabpfn_seed=TABPFN_SEED,
    device=DEVICE,

    return_predictions=True,
    verbose=True,
)


# -----------------------
# Access outputs
# -----------------------
print(results["summary"])

metrics_df = results["metrics_df"]
predictions_df = results["predictions_df"]

print("\nFold-wise metrics")
print(metrics_df)

print("\nPrediction preview")
print(predictions_df.head())

print("\nGO-LR ordering")
print(f"Learned GO-LR order length: {len(results['ordering'])}")
print("First 10 ordered features:")
print(results["ordered_feature_names"][:10])


# -----------------------
# Final 5x5 CV summary
# -----------------------
accs = metrics_df["accuracy"].to_numpy()
f1s = metrics_df["macro_f1"].to_numpy()
aucs = metrics_df["auc"].dropna().to_numpy()
```

```python
print("\nFinal 5x5 CV results")
print(f"Accuracy : {np.mean(accs):.4f} ± {np.std(accs, ddof=1):.4f}")
print(f"Macro-F1 : {np.mean(f1s):.4f} ± {np.std(f1s, ddof=1):.4f}")
print(f"AUC      : {np.mean(aucs):.4f} ± {np.std(aucs, ddof=1):.4f}")
```

```
X shape: (62, 2000), classes: 2
Using device: cuda
Learned GO-LR order length: 2000
First 10 ordered features:
['138', '1244', '1912', '1242', '381', '692', '25', '248', '697', '211']
Fold 01 Z_tr shape: (49, 64)
Fold 01: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 02 Z_tr shape: (49, 64)
Fold 02: ACC=0.8462, F1=0.8375, AUC=0.9000
Fold 03 Z_tr shape: (50, 64)
Fold 03: ACC=0.9167, F1=0.9111, AUC=1.0000
Fold 04 Z_tr shape: (50, 64)
Fold 04: ACC=0.8333, F1=0.8125, AUC=0.8438
Fold 05 Z_tr shape: (50, 64)
Fold 05: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 06 Z_tr shape: (49, 64)
Fold 06: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 07 Z_tr shape: (49, 64)
Fold 07: ACC=0.6154, F1=0.5752, AUC=0.6750
Fold 08 Z_tr shape: (50, 64)
Fold 08: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 09 Z_tr shape: (50, 64)
Fold 09: ACC=0.8333, F1=0.8125, AUC=0.9375
Fold 10 Z_tr shape: (50, 64)
Fold 10: ACC=0.9167, F1=0.8992, AUC=1.0000
Fold 11 Z_tr shape: (49, 64)
Fold 11: ACC=0.8462, F1=0.8375, AUC=0.9000
Fold 12 Z_tr shape: (49, 64)
Fold 12: ACC=0.8462, F1=0.8375, AUC=0.9250
Fold 13 Z_tr shape: (50, 64)
Fold 13: ACC=0.9167, F1=0.9111, AUC=0.9688
Fold 14 Z_tr shape: (50, 64)
Fold 14: ACC=0.8333, F1=0.8286, AUC=0.8125
Fold 15 Z_tr shape: (50, 64)
Fold 15: ACC=0.9167, F1=0.8992, AUC=0.8438
Fold 16 Z_tr shape: (49, 64)
Fold 16: ACC=0.8462, F1=0.8375, AUC=0.9000
Fold 17 Z_tr shape: (49, 64)
Fold 17: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 18 Z_tr shape: (50, 64)
Fold 18: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 19 Z_tr shape: (50, 64)
Fold 19: ACC=0.9167, F1=0.9111, AUC=1.0000
```

```
Fold 20 Z_tr shape: (50, 64)
Fold 20: ACC=0.8333, F1=0.8125, AUC=0.7500
Fold 21 Z_tr shape: (49, 64)
Fold 21: ACC=0.7692, F1=0.7451, AUC=0.8750
Fold 22 Z_tr shape: (49, 64)
Fold 22: ACC=0.6923, F1=0.6750, AUC=0.6750
Fold 23 Z_tr shape: (50, 64)
Fold 23: ACC=0.8333, F1=0.8286, AUC=0.8750
Fold 24 Z_tr shape: (50, 64)
Fold 24: ACC=0.8333, F1=0.8125, AUC=0.9688
Fold 25 Z_tr shape: (50, 64)
Fold 25: ACC=1.0000, F1=1.0000, AUC=1.0000

Final CV results
GOTabPFN completed successfully.

Task type: binary
Evaluation: 5x5 repeated stratified CV
Samples: 62
Original numeric features: 2000
GO-LR metric: euclidean
GO-LR KMeans mode: gpu
GO-LR feature subsample: full
NSC segmentation: equal_mass
NSC M rule: idf
NSC bypass_if_m_leq: 0
Device: cuda
Elapsed time: 48.30 seconds

Metric summary:
accuracy: 0.8818 ± 0.1005
macro_f1: 0.8714 ± 0.1093
auc: 0.9140 ± 0.1012
nsc_tokens: 64.0000 ± 0.0000
GOTabPFN completed successfully.
```


```
Task type: binary
Evaluation: 5x5 repeated stratified CV
Samples: 62
Original numeric features: 2000
GO-LR metric: euclidean
GO-LR KMeans mode: gpu
GO-LR feature subsample: full
NSC segmentation: equal_mass
NSC M rule: idf
NSC bypass_if_m_leq: 0
Device: cuda
Elapsed time: 48.30 seconds
```

```
Metric summary:
accuracy: 0.8818 ± 0.1005
macro_f1: 0.8714 ± 0.1093
auc: 0.9140 ± 0.1012
nsc_tokens: 64.0000 ± 0.0000

Fold-wise metrics
    fold  accuracy  macro_f1      auc  nsc_tokens
0      1  1.000000  1.000000  1.00000          64
1      2  0.846154  0.837500  0.90000          64
2      3  0.916667  0.911111  1.00000          64
3      4  0.833333  0.812500  0.84375          64
4      5  1.000000  1.000000  1.00000          64
5      6  1.000000  1.000000  1.00000          64
6      7  0.615385  0.575163  0.67500          64
7      8  1.000000  1.000000  1.00000          64
8      9  0.833333  0.812500  0.93750          64
9     10  0.916667  0.899160  1.00000          64
10    11  0.846154  0.837500  0.90000          64
11    12  0.846154  0.837500  0.92500          64
12    13  0.916667  0.911111  0.96875          64
13    14  0.833333  0.828571  0.81250          64
14    15  0.916667  0.899160  0.84375          64
15    16  0.846154  0.837500  0.90000          64
16    17  1.000000  1.000000  1.00000          64
17    18  1.000000  1.000000  1.00000          64
18    19  0.916667  0.911111  1.00000          64
19    20  0.833333  0.812500  0.75000          64
20    21  0.769231  0.745098  0.87500          64
21    22  0.692308  0.675000  0.67500          64
22    23  0.833333  0.828571  0.87500          64
23    24  0.833333  0.812500  0.96875          64
24    25  1.000000  1.000000  1.00000          64

Prediction preview
   row_index  fold true_label predicted_label
0          8     1          0               0
1         13     1          1               1
2         17     1          1               1
3         18     1          0               0
4         30     1          0               0

GO-LR ordering
Learned GO-LR order length: 2000
First 10 ordered features:
['138', '1244', '1912', '1242', '381', '692', '25', '248', '697', '211']
```

```
Final 5x5 CV results
Accuracy : 0.8818 ± 0.1005
Macro-F1 : 0.8714 ± 0.1093
AUC      : 0.9140 ± 0.1012
```

4 Colon || Binary Classification || Manual Approach

[3]:

```python
import numpy as np
import pandas as pd
import torch

from sklearn.model_selection import RepeatedStratifiedKFold
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, f1_score, roc_auc_score

from gotabpfn import GraphFeatureOrdering, pidf_segpca, TabPFN25Head,␣
↪TabPFN25Config


# -----------------------
# User settings
# -----------------------
DATA_FILE = "coloncancer_encoded.csv"  # change your dataset file name
TARGET_COL = "label"                   # change your dataset target column
SEED = 42

# Fixed GOTabPFN hyperparameters
GO_METRIC = "euclidean"
GO_NUM_CLUSTERS = 10
GO_REFINE_PASSES = 3
GO_DIRECTION_SELECT = True

NSC_SEGMENTATION = "equal_mass"
NSC_M_RULE = "idf"
NSC_TAU = 0.99
NSC_GAMMA = 1.7570143129240916
NSC_BETA = 0.2244046472232107
NSC_MMIN = 64
NSC_MMAX = 384
NSC_LMIN = 16
ASSUME_STANDARDIZED = False

TABPFN_SEED = 42
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
```

```python
# ------------------------
# Utility
# ------------------------
def compute_deltas_adjacent_corr(X_tr, Pi_star, eps=1e-12):
    """
    Compute adjacent transition scores along the GO-LR order:
        delta_t = 1 - |corr(feature_t, feature_{t+1})|.

    Required for transition-aware NSC segmentation rules:
        - equal_mass
        - largest_jump
    """
    X_t = torch.from_numpy(X_tr).float()
    perm = torch.tensor(Pi_star, dtype=torch.long)

    Xp = X_t[:, perm]
    Xc = Xp - Xp.mean(dim=0, keepdim=True)
    std = Xc.std(dim=0, unbiased=False, keepdim=True).clamp_min(eps)
    Z = Xc / std

    corr_adj = (Z[:, :-1] * Z[:, 1:]).mean(dim=0)
    deltas = 1.0 - corr_adj.abs()

    return deltas.cpu()


# ------------------------
# Load and preprocess data
# ------------------------
df = pd.read_csv(DATA_FILE)

if TARGET_COL not in df.columns:
    raise ValueError(f"TARGET_COL='{TARGET_COL}' not found in the CSV file.")

y_raw = df[TARGET_COL].astype(str).fillna("missing_target")
X_df = df.drop(columns=[TARGET_COL])

# Keep numeric features only
X_df = X_df.select_dtypes(include=[np.number])
X_df = X_df.apply(pd.to_numeric, errors="coerce")
X_df = X_df.fillna(X_df.median(numeric_only=True)).fillna(0.0)

if X_df.shape[1] == 0:
    raise ValueError("No numeric feature columns found after preprocessing.")

# Encode labels
le = LabelEncoder()
```

```python
y = le.fit_transform(y_raw).astype(np.int64)

num_classes = len(le.classes_)
if num_classes != 2:
    raise ValueError(
        f"This example expects binary classification, but found {num_classes}␣
 ↪classes."
    )

# Standardize features
scaler = StandardScaler()
X = scaler.fit_transform(X_df.values).astype(np.float32)

print(f"X shape: {X.shape}")
print(f"Classes: {list(le.classes_)}")
print(f"Using device: {DEVICE}")


# -----------------------
# Learn GO-LR feature ordering once
# -----------------------
go = GraphFeatureOrdering(
    num_clusters=GO_NUM_CLUSTERS,
    metric=GO_METRIC,
    refine=True,
    direction_select=GO_DIRECTION_SELECT,
    refine_passes=GO_REFINE_PASSES,
)

try:
    Pi_star, _, _, _ = go.fit(
        X,
        seed=SEED,
        deterministic=True,
        use_cpu_kmeans=False,
    )
except Exception:
    Pi_star, _, _, _ = go.fit(
        X,
        seed=SEED,
        deterministic=True,
        use_cpu_kmeans=True,
    )

Pi_star = list(map(int, Pi_star))

print(f"Learned GO-LR order length: {len(Pi_star)}")
```

```python
# ------------------------
# 5x5 cross-validation
# ------------------------
rkf = RepeatedStratifiedKFold(
    n_splits=5,
    n_repeats=5,
    random_state=SEED,
)

head_cfg = TabPFN25Config(
    task_type="binary",
    num_classes=2,
    device=DEVICE,
    random_state=TABPFN_SEED,
)

accs, f1s, aucs = [], [], []

for fold_id, (tr_idx, va_idx) in enumerate(rkf.split(X, y), start=1):
    X_tr, X_va = X[tr_idx], X[va_idx]
    y_tr, y_va = y[tr_idx], y[va_idx]

    nsc = pidf_segpca(
        segmentation=NSC_SEGMENTATION,
        l_min=NSC_LMIN,
        m_rule=NSC_M_RULE,
        gamma=NSC_GAMMA,
        beta=NSC_BETA,
        tau=NSC_TAU,
        M_min=NSC_MMIN,
        M_max=NSC_MMAX,
        assume_standardized=ASSUME_STANDARDIZED,
        device=DEVICE,
    )

    X_tr_t = torch.from_numpy(X_tr)

    # equal_mass and largest_jump require transition scores.
    deltas = None
    if NSC_SEGMENTATION in {"equal_mass", "largest_jump"}:
        deltas = compute_deltas_adjacent_corr(X_tr, Pi_star)

    nsc.configure(
        Pi_star=Pi_star,
        X_train=X_tr_t,
```

```
        tau=NSC_TAU,
        deltas=deltas,
    )

    Z_tr = nsc.compress(X_tr_t, mode="flatten").cpu().numpy()
    Z_va = nsc.compress(torch.from_numpy(X_va), mode="flatten").cpu().numpy()

    head = TabPFN25Head(head_cfg)
    head.fit(Z_tr, y_tr)

    P = head.predict_proba(Z_va)
    pred = np.argmax(P, axis=1)

    acc = accuracy_score(y_va, pred)
    f1 = f1_score(y_va, pred, average="macro")
    auc = roc_auc_score(y_va, P[:, 1])

    accs.append(acc)
    f1s.append(f1)
    aucs.append(auc)

    print(f"Fold {fold_id:02d}: ACC={acc:.4f}, F1={f1:.4f}, AUC={auc:.4f}")


print("\nFinal 5x5 CV results")
print(f"Accuracy : {np.mean(accs):.4f} ± {np.std(accs, ddof=1):.4f}")
print(f"Macro-F1 : {np.mean(f1s):.4f} ± {np.std(f1s, ddof=1):.4f}")
print(f"AUC      : {np.mean(aucs):.4f} ± {np.std(aucs, ddof=1):.4f}")
```

```
X shape: (62, 2000)
Classes: ['0', '1']
Using device: cuda
Learned GO-LR order length: 2000
Fold 01: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 02: ACC=0.8462, F1=0.8375, AUC=0.9000
Fold 03: ACC=0.9167, F1=0.9111, AUC=1.0000
Fold 04: ACC=0.8333, F1=0.8125, AUC=0.8438
Fold 05: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 06: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 07: ACC=0.6154, F1=0.5752, AUC=0.6750
Fold 08: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 09: ACC=0.8333, F1=0.8125, AUC=0.9375
Fold 10: ACC=0.9167, F1=0.8992, AUC=1.0000
Fold 11: ACC=0.8462, F1=0.8375, AUC=0.9000
Fold 12: ACC=0.8462, F1=0.8375, AUC=0.9250
Fold 13: ACC=0.9167, F1=0.9111, AUC=0.9688
Fold 14: ACC=0.8333, F1=0.8286, AUC=0.8125
Fold 15: ACC=0.9167, F1=0.8992, AUC=0.8438
```

```
Fold 16: ACC=0.8462, F1=0.8375, AUC=0.9000
Fold 17: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 18: ACC=1.0000, F1=1.0000, AUC=1.0000
Fold 19: ACC=0.9167, F1=0.9111, AUC=1.0000
Fold 20: ACC=0.8333, F1=0.8125, AUC=0.7500
Fold 21: ACC=0.7692, F1=0.7451, AUC=0.8750
Fold 22: ACC=0.6923, F1=0.6750, AUC=0.6750
Fold 23: ACC=0.8333, F1=0.8286, AUC=0.8750
Fold 24: ACC=0.8333, F1=0.8125, AUC=0.9688
Fold 25: ACC=1.0000, F1=1.0000, AUC=1.0000

Final 5x5 CV results
Accuracy : 0.8818 ± 0.1005
Macro-F1 : 0.8714 ± 0.1093
AUC      : 0.9140 ± 0.1012
```

[4]:
```python
import gotabpfn
import tabpfn
import inspect

print("gotabpfn version:", gotabpfn.__version__)
print("gotabpfn path:", gotabpfn.__file__)
print("run_gotabpfn_csv file:", inspect.getsourcefile(gotabpfn.
 ↪run_gotabpfn_csv))

print("tabpfn version:", getattr(tabpfn, "__version__", "NO __version__"))
print("tabpfn path:", tabpfn.__file__)
```

```
gotabpfn version: 0.1.11
gotabpfn path: /usr/local/lib/python3.12/dist-packages/gotabpfn/__init__.py
run_gotabpfn_csv file: /usr/local/lib/python3.12/dist-
packages/gotabpfn/gotabpfn.py
tabpfn version: 6.3.1
tabpfn path: /usr/local/lib/python3.12/dist-packages/tabpfn/__init__.py
```

[ ]:
```python

```